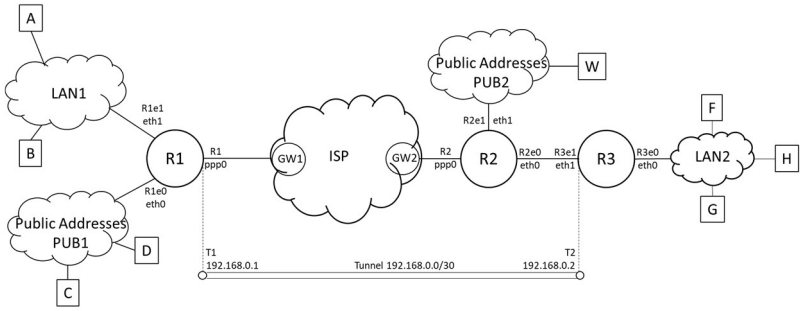
Examen final. Xarxes de Computadors (XC), Grau en Enginyeria Informàtica	19/06/2025	Spring 2025	Code 4

All questions have a single true answer. All are scored *p*, −*p*/2 if wrong, 0 if no answer.

Problem IP (2.5 points)

The figure shows the network of an organization and its connection to the Internet. Each router interface is labelled with its IP address and interface name. Hosts and servers are labelled in uppercase the IP address and in lowercase the MAC addresses. Server with IP address W and MAC address w is the organization's web server. The range of public IP addresses assigned is 100.100.112.0/21. Private addresses use the range 10.10.0.0/16. The internal network with LAN1 and LAN2 uses private addresses. That is, A, B, F, G and H have private addresses.

The DNS is provided by the ISP (Internet Service Provider). Traffic between LAN1 and LAN2 is routed through the tunnel. All clients have Internet access. The Firewall is located in R1 and R3.



The broadcast address corresponding to the bloc of public IP addresses is:

1. A 100.100.119.255 B 100.100.112.255 C 100.100.127.255 D 100.100.255.255

Network PUB1 has a /23 mask; the rest of available public addresses is allocated as follows: first, the smallest possible network to link R2-R3, and the rest is assigned to PUB2 so that it may allocate the maximum number of devices. Assign the public IP addresses in this order (smallest to highest IP): PUB1, R2-R3 and PUB2. Identify the correct Ip addresses of the networks and interfaces shown below.

2. PUB1: A 100.100.114.0/22 B 100.100.112.0/21 C 100.100.112.0/30 D 100.100.112.0/23
3. BcastPUB1: A 100.100.255.255/23 B 100.100.113.255/23 C 100.100.112.255/23 D 100.100.114.255/23
4. R2-R3: A 100.100.120.0/30 B 100.100.116.0/30 C 100.100.114.0/30 D 100.100.112.0/30
5. R3e1: A 100.100.114.3/30 B 100.100.114.2/30 C 100.100.116.1/30 D 100.100.120.1/30
6. R2e1: A 100.100.114.1/22 B 100.100.114.1/20 C 100.100.116.1/20 D 100.100.116.1/22

Private addresses (10.10.0.0/16) are distributed to LAN1 and LAN2 starting by LAN1. Distribute the available addresses so that LAN2 has the maximum number of addresses. Assign IP addresses to the following networks and interfaces.

7. LAN1: A 10.10.0.0/16 B 10.10.1.0/17 C 10.10.0.0/17 D 10.10.128.0/16
8. Bcast1: A 10.10.127.255/17 B 10.10.0.255/17 C 10.10.255.255/16 D 10.10.1.255/16
9. LAN2: A 10.10.128.0/17 B 10.10.192.0/17 C 10.10.64.0/18 D 10.10.64.0/20
10. Bcast2: A 10.10.255.255/17 B 10.10.192.255/17 C 10.10.127.255/18 D 10.10.112.255/20

Which interface should apply PNAT (*Port and Address Translation*)?

11. R1: A ppp0 B eth1 C eth0 D cap
12. R2: A ppp0 B eth1 C eth2 D cap
13. R3: A T2 B eth0 C eth1 D cap

Complete the routing tables for R1, R2 and R3 using the notation shown in the figure for IP addresses. Consider the tunnel and Internet access.

R1

network	Gw	Iface
LAN1	---	eth1
(14)	(14)	eth0
GW1/32	---	ppp0
192.168.0.0/30	---	tun0
LAN2	T2	tun0
(15)	(15)	ppp0

R2

network	Gw	Iface
PUB2	---	eth1
R2-R3	---	eth0
GW2/32	---	ppp0
(16)	(16)	(16)

R3

network	Gw	Iface
LAN2	---	eth0
R3-R2	---	eth1
(18)	(18)	tun0
(19)	(19)	tun0
0.0.0.0/0	R2	eth1

14. Cells 14: A PUB1, --- B 0.0.0.0/0, --- C PUB2, T2 D C, ---
15. Cells 15: A R2ppp0, GW1 B ISP-GW1 – GW1 C LAN2, T2 D 0.0.0.0/0, GW1
16. Cells 16: A 0.0.0.0/0, GW2, ppp0 B PUB1, GW2, ppp0 C LAN1, T2, eth0 D LAN2, R3e1, eth0
17. Cells 18: A 192.168.16.0/24 B 192.168.0.0/30, --- C LAN1, R1ppp0 D R2e0, ---
18. Cells 19: A LAN1, T1 B PUB1, T1 C PUB1, T2 D LAN1, T2

Initially, all ARP tables are empty, except the ones at interfaces R1ppp0 and R2ppp0. Complete the contents of the ARP tables at the following interfaces if device A executes the command "*ping www.trademark.org*". This web server is W in the figure.

19. A: A R1e1, r1e1 B W, w C R1, ppp0 D cap
20. R1e1: A GW, gw B W, w C A, a D cap
21. R1e0: A GW1, gw1 B A, a C R1, ppp0 D cap
22. R2e0: A GW2, gw2 B A, a C W, w D cap
23. R2e1: A W, w B A, a C R2, ppp0 D cap

In the same case, how many ARP messages, ICMP and UDP(DNS) datagrams go through interface R1 (R1ppp0) until the first response comes back?

24. A 0, 4, 1 B 2, 2, 2 C 0, 2, 2 D 1, 1, 2

In the same case, what are the source and destination IP addresses for the datagrams reaching W?

25. IPs: A R1ppp0 - W B A - W C R1e1 -W D T1 - W

In the same case, what are the source and destination IP addresses for the datagrams entering interface R2ppp0?

26. IPs: A R1ppp0 - W B A - W C R1ppp0 – R2ppp0 D T1 -W

Device A executes the command "*traceroute H*". What IP addresses will appear in the output of the *traceroute*?

27. IP: A GW1 B R1ppp0 C T1 D R1e1
28. IP: A R3e0 B R2e0 C H D R2ppp0

In the same case, what are the source and destination IP addresses for the datagrams entering interface R2ppp0?

29. IPs: A A - H B R1ppp0 – R3e1 C T1 - H D R1ppp0 – T2

In the same case, what are the source and destination IP addresses for the datagrams sent from interface R3e0?

30. IPs: A A - H B R1ppp0 - H C T1 - H D A – T2

Examen final de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	19/06/2025	Primavera 2025

Duration: 2h45m. All questions have a single true answer. All are scored p, $-p/3$ if wrong, 0 if no answer.

QUIZ (2.5 punts)

In the ISO OSI reference model, the following devices must have the following levels:

1. Router: A. 1 B. 1,2 C. 1,2,3 D. 1,2,3,4
2. Switch: A. 1 B. 1,2,3,4 C. 1,2,3 D. 1,2
3. Hub: A. 1,2 B. 1,2,3 C. 1 D. 1,2,3,4

Say which fields can be found in the messages of the following protocols:

4. Protocol field: A. DHCP B. ARP C. TCP D. IP
5. MAC address: A. ARP B. TCP C. IP D. DHCP
6. Gateway: A. DHCP B. IP C. TCP D. RIP

Say what characteristics may correspond to the following protocols

7. Persistent: A. mail B. TCP C. RIP D. HTTP
8. Reliable: A. TCP B. UDP C. DHCP D. IP
9. Connection oriented: A. IP B. all false C. UDP D. SMTP

Say which class the following IP addresses correspond to :

10. 127.0.0.1: A. class B B. class D C. class A D. class C
11. 224.0.0.1: A. class A B. class D C. class C D. class B
12. 190.0.0.1: A. class A B. class D C. class C D. class B

#	D	Gw	M	If
1	0.0.0.0/0	R1	2	e0
2	1.0.0.0/16	R2	3	e1
3	200.0.1.0/25	-	1	e0
4	200.0.1.128/25	-	1	e1

In a network with RIPv2 with split-horizon and class summarization, one of the routers has the RT in the figure. Say what update messages it will send through the indicated interfaces. Notation: $\{(D; M), \cdots\}$

13. e0: A. {(0.0.0.0/0; 2), (1.0.0.0/16; 3), (200.0.1.128/25; 1)} B. {(1.0.0.0/8; 3), (200.0.1.0/24; 1)} C. {(0.0.0.0/0; 2), (1.0.0.0/16; 3), (200.0.1.0/25; 1), (200.0.1.128/25; 1)} D. {(1.0.0.0/8; 3), (200.0.1.128/25; 1)}
14. e1: A. {(0.0.0.0/0; 2), (1.0.0.0/16; 3), (200.0.1.128/25; 1)} B. {(0.0.0.0/0; 2), (200.0.1.0/25; 1)} C. {(1.0.0.0/8; 3), (200.0.1.0/24; 1)} D. {(0.0.0.0/0; 2), (1.0.0.0/16; 3), (200.0.1.0/25; 1), (200.0.1.128/25; 1)}

If the router with the RT in the figure receives an update message with the indicated tuple by e0 from router R1, tell what changes will occur in the RT. Notation: update (D; M), TE: {D; Gw; M; If}

15. (1.1.0.0/24; 2): A. it will change #2 by {1.0.0.0/8; R1; 3; e0} B. it will add {1.1.0.0/24; R1; 2; e0} C. it will add {1.1.0.0/24; R1; 3; e0} D. all false
16. (0.0.0.0/0; 16): A. all false B. it will change #1 by {0.0.0.0/0; R1; 3; e0} C. it will change #1 by {0.0.0.0/0; R1; 17; e0} D. it will change #1 by {0.0.0.0/0; R1; 16; e0}

Say me what the prefix will be that aggregates the following networks with the largest mask :

17. 80.80.128.0/19, 80.80.168.0/21: A. 80.80.128.0/18 B. 80.80.160.0/19 C. 80.80.0.0/16 D. 80.80.128.0/17
18. 10.0.64.0/18, 10.0.32.0/19, 10.0.16.0/20: A. 10.0.0.0/17 B. 10.0.0.0/16 C. 10.0.0.0/18 D. 10.0.64.0/18

Which of the following subnets are valid subnets of the network 1.0.0.0/18?

19. A. 1.0.0.0/20, 1.0.8.0/21 B. 1.0.0.0/20, 1.0.16.0/21 C. 1.0.0.0/20, 1.0.0.0/21 D. 1.0.0.0/19, 1.0.16.0/19
20. A. 1.0.0.0/30, 1.0.0.2/30 B. 1.0.0.0/20, 1.0.0.4/30 C. 1.0.0.0/30, 1.0.0.4/30 D. 1.0.0.2/30, 1.0.4.0/30

The ACL in the figure is applied to datagrams entering the interface that connects a router to the ISP. All IPs on the internal network are public. N1 is one of the internal networks, S is a DMZ server. The ports are: 53 DNS, 80 web, 1194 openvpn. There is a default rule "drop all". Notation: source $\rightarrow$ destination of the connection. Say which connections (with traffic in both directions) are possible (all IPs answer) :

Protocol	IP_{src}	$Port_{src}$	IP_{dst}	$Port_{dst}$	Acció
UDP	any	53	S	≥ 1024	Permit
TCP	any	≥ 1024	S	1194	Permit
any	any	-	S	-	Deny
TCP|UDP	any	< 1024	N1	≥ 1024	Permit
ICMP	any	-	any	-	Permit

21. A. Internet $\rightarrow$ NS from S B. N1 $\rightarrow$ NS from the Internet C. Internet $\rightarrow$ NS from N1 D. all false
22. A. all false B. Internet $\rightarrow$ openvpn from S C. Internet $\rightarrow$ web from S D. Internet $\rightarrow$ ping S
23. A. all false B. S $\rightarrow$ openvpn from the Internet C. S $\rightarrow$ ping Internet D. N1 $\rightarrow$ openvpn from the Internet

In a TCP connection with an RTT=200ms and throughput of 10 Mbps :

24. The average window is approx.: A. 65 kB B. 125 kB C. all false D. 250 kB
25. During 1 minute are sent approx.: A. 100 MB B. 600 MB C. all false D. 75 MB

Say which resource records have the following information in the data field

26. IP address: A. CNAME B. MX C. all false D. NS
27. Priority: A. all false B. CNAME C. NS D. MX

Say which statements are true about the following protocols :

28. UDP: A. has a fixed size of 20 bytes B. has a DF flag C. has a checksum field D. all false
29. SMTP: A. one of the commands sent by the client is HELO B. one of the fields can be content-transfer-encoding C. all false D. has a checksum field
30. RIP: A. all false B. can summarize networks 10.0.0.0/25 and 10.0.0.128/25 with 10.0.0.0/24 C. can detect if a neighboring router hangs D. may converge to non-optimal routing tables (non-minimum metrics)
31. HTTP: A. the server can send an error code in the status line B. the server can send an error code in one of the header fields C. the client can send an error code in the request line D. all false

About the basic TCP protocol with SS/CA

32. A. the "identification" field in the header identifies fragments of the same segment B. all false C. the "advertised window" header field varies depending on network congestion D. the "congestion window" field in the header varies depending on network congestion
33. A. the advertised window awnd only increases when acks confirming new data arrive B. in the "slow start" phase the window increases more slowly than in the "congestion avoidance" phase C. all false D. the congestion window cwnd only increases when acks confirming new data arrive

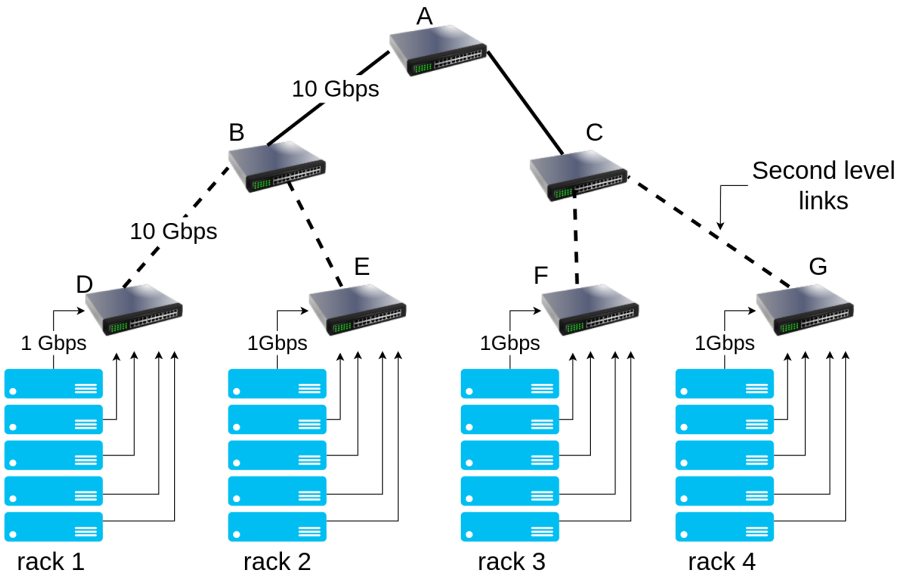
We want a router with a 100 Mpbs transmission line to introduce a maximum delay of 20 ms

34. The buffer size must be at most: A. 2000 kB B. 250 kB C. 500 kB D. all false

Examen final de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	19/06/2025	Primavera 2025

Duration: 2h45m. All questions have a single true answer. All are scored p, $-p/3$ if wrong, 0 if no answer.

Problem LAN (1,5 pts)



The figure shows a small data center and its network, consisting of 4 racks with 5 servers each, and a network of 7 switches that interconnects them. Each server has a 1 Gbps Ethernet connection directly to the switch in its rack, and the links between the switches are all 10 Gbps. Assume that: (1) All links are full duplex, (2) There are only the data flows indicated in each question, (3) The traffic generated by return acks is considered negligible for the purposes of estimating the throughput of a link, (4) There are no VLANs configured unless specifically stated.

Consider switch A. What is its level 2 (Ethernet) broadcast domain?

1. A. Only the link switch A - switch B B. All the links connected to switch A C. All the network D. All the network, except the servers

Is it possible that switch C's MAC address table has one of the MAC addresses of servers in rack 1?

2. A. Yes B. Only if we add it manually C. Only if VLANs are configured D. No

Suppose each of the servers in rack 1 sends data at the highest possible speed to the five servers in rack 2, that is, five data streams per server, for each server. What will be the throughput **of each individual data flow**?

3. A. 200 Mbps B. 0,5 Gbps C. 1 Gbps D. 100 Mbps

In the same scenario as above, what will be the traffic in the link between switch B and switch E?

4. A. 1 Gbps B. 2 Gbps C. 5 Gbps D. 10 Gbps

Continuing with the previous scenario (all servers in rack 1 sending data to all servers in rack 2), now, in addition, one server in rack 2 sends data at maximum speed to another server in rack 1. What will be the throughput of this data transmission?

5. A. 200 Mbps B. 100 Mbps C. 0,5 Gbps D. 1 Gbps

How many additional servers with the same characteristics should we add in rack 1 to saturate the link switch B - switch D?

6. A. 2 B. 6 C. 5 D. 4

Due to cost restrictions, we decide to change all the links in the second level (marked with a dashed line in the diagram) to 1 Gbps speed. Suppose that the five servers in rack 1 are all sending data at the maximum possible speed to **the same server in rack 2**. What will be the throughput of each server in rack 1 now?

7. A. 500 Mbps B. 200 Mbps C. 100 Mbps D. 1 Gbps

Continuing with the previous scenario, what will be the throughput in the server in rack 2 that receives the connections, the five incoming connections in total?

8. A. 1 Gbps B. 100 Mbps C. 200 Mbps D. 500 Mbps

Continuing with the scenario from question 7 (second-level links at 1 Gbps), now all servers in rack 1 are sending data at the maximum possible speed **to all servers in the other three racks**. Which link will be the bottleneck?

9. A. Link switch D - switch B B. Link switch A - switch B C. Links servers rack 3 - switch F D. Links servers rack 1 - switch D

Next, related to the scenario in the previous question, what will be the throughput of **each individual connection** from a server in rack 1? Keep in mind that each server talks to all the servers in the other three racks.

10. A. 66,6 Mbps B. 13,3 Mbps C. 200 Mbps D. 2,5 Mbps

Suppose a new wire is connected between switch B and C. What will happen?

11. A. The Spanning Tree Protocol (STP) will turn off one of the links B. Nothing C. STP will send an error message D. STP will leave all links active

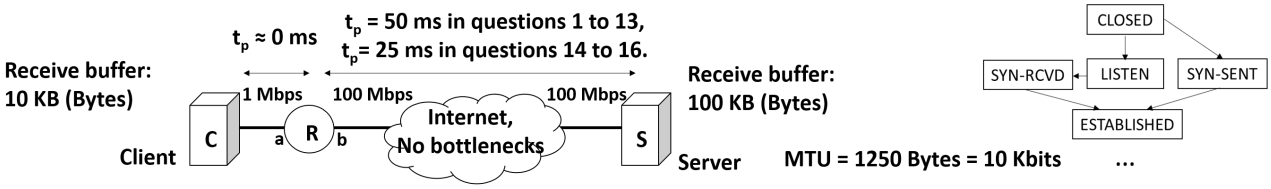
We want to create different VLANs, as some servers are from different customers. Since we do not know which server will be assigned to each client, we will dynamically assign them to the corresponding VLAN. For example, a rack 1 server and one from rack 2 may have the same VLAN. Given this, which links will need to be set up in Trunk mode, assuming that it can also happen that a rack 1 server has to talk to one of rack 4?

12. A. All links between servers and switches (e.g. switch D - rack 1) B. Only links switch A - switch B and switch A - switch C C. All links between switches, e.g. switch B - switch D D. All links

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TCP Problem (2,5 points)



In the network shown in the figure, we assume that, at time t = 0, a web browser in C establishes a TCP connection with server S, on which a web server process runs. Unless stated otherwise, we assume that no packets are lost. Neglect the size of the IP and TCP headers. The time *tp* refers to the propagation times in the Internet section of the connection (one way). We assume an uncongested Internet that can support connections up to 100 Mbps without introducing further delays. When computing times, neglect transmission times. Always choose the answer that you think is closest to the exact answer ("all false" means that all are far from the exact answer). awnd refers to the *advertised window*, cnwd to the *congestion window*, and wnd=$\min$(cwnd, awnd) to the *tcp window*.

What will the states of the TCP sockets opened by the web browser in C and the web server in S be?

1. t=25 ms: A. C: LISTEN B. S: CLOSED C. C: SYN-SENT D. S: SYN-SENT
2. t= 75 ms: A. C: SYN-RCVD B. C: SYN-SENT and LISTEN C. S: LISTEN D. S: SYN-RCVD and LISTEN
3. t= 175 ms: A. C: SYN-SENT B. S: SYN-SENT C. S: ESTABLISHED and LISTEN D. C: ESTABLISHED and LISTEN

What would the values of awnd and window-scale (w-s) be? (assume TCP uses this option, w-s is the exponent of the *window scaling factor*)

4. SYN packet: A. awnd=50 KB, w-s=1 B. awnd=10 KB, w-s=1 C. awnd=100 KB, w-s=0 D. awnd=10 KB, w-s=0
5. SYN+ACK packet: A. awnd=50 KB, w-s=1 B. awnd=10 KB, w-s=0 C. awnd=100 KB, w-s=0 D. awnd=10 KB, w-s=1

Assume that once the client opens the TCP connection, a download a 12.5 KB (100 Kb) file from the server starts. What would the download time be for this file as seen by the client? We count the time from when the client opens the TCP connection until the file is completely received. We do not include the time required for closing the TCP connection.

6. A. 500 ms B. 250 ms C. 400 ms D. 600 ms

What would the values of the congestion window (cwnd) and slow start threshold (ssthr) be at the server at t= 300 ms?

7. A. cwnd= 2.5 KB, ssthr= 25 KB B. cwnd= 1.25 KB, ssthr= ∞ C. all false D. cwnd= 2.5 KB, ssthr= ∞

Assume that once the client completes the transfer of the first file, it sends a request to download a second file of size 5 KB (40 Kb) from the server S. *The download of the second file is done using the same TCP connection*. What would be the download time for the second file? We count the download time from when the client sends a download request for the the second file (just after the entire first file has been downloaded) to when the client finishes receiving the second file.

8. A. 100 ms B. all false C. 50 ms D. 200 ms

Previously, we assumed that the process reading the TCP receive buffer at C extracted information from the buffer as soon as data arrived. Now, however, we will assume that the process reading the receive buffer at C froze after the 3WH was completed. What will the awnd be in the ACK packet that acknowledges the TCP segment carrying the bytes [3750, 5000) of *the first file*?

9. A. all false B. awnd=8.75 KB C. awnd=96 KB D. awnd=5 KB

Given the assumption of the previous question, what will the cwnd be at the server after it receives the ACK of the TCP segment carrying the bytes [3750, 5000) of the first file?

10. A. all false B. cwnd= 2.5 KB C. cwnd=6.25 KB D. cwnd= 7.5 KB

Now, assume that everything goes back to normal at C, the previous connection is completed, and C closes the TCP connection. Then, C starts a new TCP connection to S, with the same receive buffers size at C and S as before, and downloads a very large file (i.e. a bulk transfer scenario). What would the average throughput, Th, be for this transfer (assuming no losses)?

11. A. Th= 1 Mbps B. Th=800 Kbps C. Th= 500 Kbps D. Th= 400 Kpbs

Given the assumptions of the previous question. What would be the average occupation, Q, at the output buffers of the interface "a" of router R, assuming no other connections are present?

12. A. Q approx 10 KB B. Q approx 5 KB C. all false D. Q approx 0

Given the assumptions of the previous question: What would the tcp window, wnd$=\min$(cwnd,awnd), and ssthr be in S at the end of the transfer (we assume a very large file transfer and no losses)?

13. A. wnd=100 KB, ssthr=∞ B. all false C. wnd=∞, ssthr=∞ D. wnd=10 KB, ssthr=∞

We keep the assumptions of the previous question, with a single change: now the propagation time in the Internet part of the connection is reduced to tp=25 ms. What would the average throughput, and the average occupation at the output buffers of the interface "a" of router R, be for this transfer (assuming no losses and no other connections are present)?

14. A. Th= 500 Kbps, Q approx 5 KB B. Th= 1 Mbps, Q approx 3.75 KB C. Th= 800 Kbps, Q approx 5 KB D. all false

After wnd=min(cwnd,awnd) reaches its maximum value, we have a loss in one packet (no more losses occur). What would be the tcp window, wnd, and ssthr at the Server at the end of the transfer?

15. A. wnd =10 KB, ssthr = 10 KB B. wnd =100 KB, ssthr = ∞ C. all false D. wnd =50 KB, ssthr = ∞

Given the assumptions of the previous question. What will be the state of the congestion control mechanism in C and S at the end of the transfer?

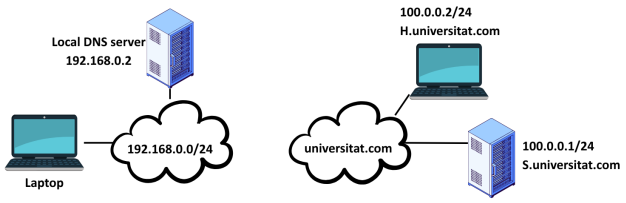
16. A. Client: CA, Server: SS B. Client: CA, Server: CA C. Client: SS, Server: SS D. Client: SS, Server: CA

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Problem App (1 point)

We want to configure the name server (authority) for the domain *universitat.com* in the network shown in the figure. On server S there is the name server, web (canonic name), and mail with names ns, www, mail, and mail2, respectively. It must be configured to try the connection with mail first before with mail2. The laptop in the private network uses a local DNS server at 192.168.0.2 for DNS resolution. This local DNS server is responsible for resolving external domains, such as *universitat.com*.



Which of the following Resource Records will be in the server configuration file?

1. A. S.universitat.com A 100.0.0.1 B. universitat.com CNAME ns.universitat.com C. www.universitat.com A 100.0.0.1 D. H.universitat.com CNAME S.universitat.com
2. A. universitat.com MX 10 mail.universitat.com; universitat.com MX 20 mail2.universitat.com B. universitat.com MX 10 mail.universitat.com; universitat.com A 100.0.0.1 C. universitat.com CNAME mail.universitat.com; universitat.com A 100.0.0.1 D. universitat.com MX 20 mail.universitat.com; universitat.com MX 10 mail2.universitat.com

A user accesses *www.universitat.com* from a browser in the laptop of the private network. How many different name servers will the local DNS server contact to fully resolve the URL? Assume the **local DNS server does not have any cached entry**.

3. A. 2 B. 3 C. 1 D. 4

User *a@universitat.com* sends an email to *b@universitat.com* with the subject *"Dubte XC"*, content *"Bon dia, tinc un dubte sobre la capçalera IP"* and an attached image *"captura.png"*.

Which line of the message is necessary to allow the sending of the text and attached image?

4. A. content-type: multipart/alternative B. content-type: image/png C. content-type: text/plain; charset=iso-8859-1 D. content-type: multipart/mixed; boundary=BB

What combination of MIME headers should be used to correctly send the text content?

5. A. Content-Type: text/plain; charset=UTF-8; Content-Transfer-Encoding: quoted-printable B. Content-Type: text/html; charset=US-ASCII; Content-Transfer-Encoding: quoted-printable C. Content-Type: text/plain; Content-Transfer-Encoding: base64 D. Content-Type: text/plain; charset=US-UTF-8; Content-Transfer-Encoding: 7bit

In addition to DNS and email services, the domain universitat.com serves web pages located in the same server. The page www.universitat.com returns an HTML file including 3 images. The complete DNS resolution takes 40 ms, and the RTT between client and server is also 40 ms. HTTP data transfer time per object (web page and images) is 10 ms.

Which is the correct order of protocol messages exchanged between client and server when downloading www.universitat.com? Assume **no DNS caching** (the browser doesn't have the IP addresses).

6. A. TCP handshake → HTTP request → DNS query B. DNS query → HTTP request → TCP handshake C. HTTP request → DNS query → TCP handshake D. DNS query → TCP handshake → HTTP request(s)

How long does it take to download www.universitat.com **without images**? Assume there is **not DNS caching** (the browser doesn't have the IP addresses) and a **single HTTP/1.1 connection without pipelining**.

7. A. 120 ms B. 90 ms C. 160 ms D. 130 ms

How long does it take to download www.universitat.com with the **3 images**? Assume there is **not DNS caching** (the browser doesn't have the IP addresses) and a **single HTTP/1.1 connection with pipelining**.

8. A. 200 ms B. 160 ms C. 180 ms D. 240 ms

How long does it take to download www.universitat.com with the **3 images**? Assume there is **DNS caching** (the browser has the IP addresses) and a **single HTTP/1.1 connection with pipelining**.

9. A. 140 ms B. 160 ms C. 180 ms D. 120 ms

How long does it take to download www.universitat.com with the **3 images**? Assume there is **not DNS caching** (the browser doesn't have the IP addresses) and a **single HTTP/1.1 connection without pipelining**.

10. A. 280 ms B. 200 ms C. 180 ms D. 240 ms

How long does it take to download www.universitat.com with the 3 images if the browser opens 3 parallel TCP connections to download the images? Assume there is **DNS caching** (the browser has the IP addresses) and the browser must **first download the HTML file** to discover and open the parallel TCP connections to request the additional objects (such as the images) referenced within the page.

11. A. 160 ms B. 120 ms C. 150 ms D. 180 ms